

| #                             | Category | Item / Function                     |
|-------------------------------|----------|-------------------------------------|
| <b>MOTOR &amp; DRIVE</b>      |          |                                     |
| 1                             |          | AC Motor (bench test unit)          |
|                               |          | VFD (Variable Frequency Drive)      |
| 2                             |          |                                     |
|                               |          | Dynamic Braking Resistor            |
| 3                             |          |                                     |
| <b>POWER DISTRIBUTION</b>     |          |                                     |
| 4                             |          | Main Power Switch                   |
|                               |          | VFD Branch Breaker                  |
| 5                             |          |                                     |
|                               |          | DC Power Supply                     |
| 6                             |          |                                     |
|                               |          | Main Power Contactor                |
| 7                             |          |                                     |
| 8                             |          | Fuse Block - 24V Button Common      |
| 9                             |          | Fuse Block - AC Branches            |
| <b>PLC CONTROL SYSTEM</b>     |          |                                     |
| 10                            |          | PLC CPU / Ethernet Module           |
| 11                            |          | PLC Rack / Base                     |
| 12                            |          | Discrete Input Module (24 VDC)      |
| 13                            |          | Analog Current Input Module         |
|                               |          | RTD Input Module                    |
| 14                            |          |                                     |
|                               |          | Relay Output Module                 |
| 15                            |          |                                     |
|                               |          | Analog Voltage Output Module        |
| 16                            |          |                                     |
| 17                            |          | PLC Rack Filler Panels              |
| <b>HMI / OPERATOR DISPLAY</b> |          |                                     |
| 18                            |          | C-More Micro Touch Panel            |
| <b>OPERATOR CONTROLS</b>      |          |                                     |
| 19                            |          | Forward Pushbutton Head             |
| 20                            |          | Reverse Pushbutton Head             |
| 21                            |          | Deadman/Run Pushbutton Head         |
| 22                            |          | Reset Pushbutton Head               |
|                               |          | NO Contact Block (1NO finger-safe)  |
| 23                            |          |                                     |
| 24                            |          | LED Block - Green (24V DC)          |
| 25                            |          | LED Block - Yellow (24V DC)         |
| 26                            |          | LED Block - Blue (24V DC)           |
| 27                            |          | LED Block - Red (24V DC)            |
|                               |          | Emergency Stop Pushbutton           |
| 28                            |          |                                     |
| 29                            |          | Panel Indicator Light - Error/Fault |
| <b>MOTOR CONNECTIONS</b>      |          |                                     |
| 30                            |          | Banana Socket - Blue (4 mm)         |

- 31 Banana Socket - White (4 mm)
- 32 Banana Socket - Red (4 mm)

#### **WIRE, DUCT, & HARDWARE**

- 33 Slotted Wire Duct (Type G, 1" x 1")
- 34 Duct Cover for G1X1LG6
- 35 Insulated Crimp Ferrules, 12 AWG (gray)
- 36 Control Wire - 18 AWG stranded Blue
- 37 Return Wire - 18 AWG White or Blue/W
- 38 Power Wire - 12 AWG Black
- 39 Ground Wire - 12 AWG Green/Yellow
- Fuse - 500 mA fast-blow (5x20 mm)
- 40
- 41 Fuse - 1 A fast-blow (5x20 mm)

| Manufacturer          | Model / Part Number        | Qty | Unit |
|-----------------------|----------------------------|-----|------|
| GE                    | 5KE182BC205B               | 1   | EA   |
| Huanyang              | HY02D211B-T                | 1   | EA   |
| ATO                   | APCS-300R30-AD             | 1   | EA   |
| Powertec              | 71008                      | 1   | EA   |
| Phoenix Contact       | TMC 8 3C 15A (P/N 2907618) | 1   | EA   |
| Mean Well             | NDR-480-24                 | 1   | EA   |
| Mitsubishi            | SD-N35                     | 1   | EA   |
| Morsettitalia         | EURO S4LH                  | 1   | EA   |
| Morsettitalia         | EURO S10H-5H               | 1   | EA   |
| AutomationDirect      | H2-DM1E                    | 1   | EA   |
| AutomationDirect      | D2-09B-1                   | 1   | EA   |
| AutomationDirect      | D2-08ND3                   | 1   | EA   |
| AutomationDirect      | F2-08AD-1                  | 1   | EA   |
| AutomationDirect      | F2-04RTD                   | 1   | EA   |
| AutomationDirect      | D2-08TR                    | 1   | EA   |
| AutomationDirect      | F2-08DA-2                  | 1   | EA   |
| AutomationDirect      | D2-Fill                    | 3   | EA   |
| AutomationDirect      | EA1-T4CL                   | 1   | EA   |
| Autonics              | S2PR-P3G                   | 1   | EA   |
| Autonics              | S2PR-P3Y                   | 1   | EA   |
| Autonics              | S2PR-P3B                   | 2   | EA   |
| Autonics              | S2PR-P3R                   | 1   | EA   |
| Autonics              | SA-CA                      | 5   | EA   |
| Autonics              | SA-LDG                     | 1   | EA   |
| Autonics              | SA-LDY                     | 1   | EA   |
| Autonics              | SA-LDB                     | 2   | EA   |
| Autonics              | SA-LDR                     | 1   | EA   |
| Cutler-Hammer / Eaton | E22B1                      | 1   | EA   |
| Wamco                 | 525                        | 1   | EA   |
| Generic               | Banana Socket Blue         | 1   | EA   |

|                        |                          |        |
|------------------------|--------------------------|--------|
| Generic                | Banana Socket White      | 1 EA   |
| Generic                | Banana Socket Red        | 1 EA   |
|                        |                          |        |
| Panduit                | G1X1LG6                  | 6 FT   |
| Panduit                | C1LG6                    | 6 FT   |
| Z+F                    | V30AE000055              | 500 EA |
| Various                | 18 AWG Blue stranded     | 1 Roll |
| Various                | 18 AWG White stranded    | 1 Roll |
| Various                | 12 AWG Black             | 1 Roll |
| Various                | 12 AWG Green/Yellow      | 1 Roll |
| Bussmann or Littelfuse | GMA-500mA or 0217.500MXP | 1 EA   |
| Bussmann or Littelfuse | GMA-1A or equiv.         | 1 EA   |

## Notes / Source

3 HP, 230V 1750 RPM 4-pole TEFC. See motor/Motor\_Nameplate.md.

2.2 kW (3 HP), 110V single-phase in, 220V 3-phase out. See vfd/Programmed\_Parameters.md.

300 W, 30 ohm, aluminum-clad. P+/PR terminals on VFD. See vfd/Brake\_Resistor\_Wiring.md.

120 V ON/OFF disconnect. Powers entire control panel.

15 A thermal-magnetic DIN-rail breaker on VFD neutral leg (per as-built; reviewing for final).

480 W, 24 VDC, DIN-rail. Powers PLC, HMI, indicator LEDs, VFD control loops.

3-pole magnetic contactor. Energized through E-stop chain via E22B1 NC.

Holds 1 A fast-blow fuse for shared pushbutton +24V supply.

Holds AC line branch fuses for PSU and PLC mains feeds.

Do-more CPU with built-in Ethernet, fits DL205 rack.

9-slot DL205 base, AC powered.

8-ch DI, sourcing mode. Slot 2. Inputs X0-X7.

8-ch analog input. Slot 3. CH1+ reads VFD VO motor current monitor.

4-ch RTD input. Slot 4. Installed but unused (spare for future thermal sensing).

8-ch relay output. Slot 5. Sinking mode. Drives button LEDs (Y0-Y4) and VFD FOR/REV/RST (Y6/Y7/Y5).

8-ch voltage output. Slot 6. +V1 drives VFD VI speed reference (0-10V).

Slots 7-9 fillers.

4-inch color touch panel HMI. Ethernet to PLC.

22 mm illuminated green pushbutton (Forward command -> X0).

22 mm illuminated yellow pushbutton (Reverse command -> X1).

22 mm illuminated blue pushbuttons (Run 1 -> X2, Run 2 -> X3).

22 mm illuminated red pushbutton (planned Reset -> X6).

One per pushbutton (was MC22-UA legacy P/N). 110VAC/10A, 220VAC/5A.

Forward button illumination -> Y0.

Reverse button illumination -> Y1.

Run 1 and Run 2 button illumination -> Y2, Y3.

Reset button illumination (planned).

22 mm mushroom E-stop, NC contact. In series with SD-N35 contactor coil.

28 VDC, 0.6 W panel-mount LED indicator. Driven by Y4.

Quick-disconnect for VFD U output to motor T1.

VFD V output to motor T2.

VFD W output to motor T3.

Light gray, 6 ft lengths. Panel wire management.

Light gray, snap-on cover.

10 mm length, DIN color code. Per pack.

DC control circuits per NFPA 79 §13.2.

DC return circuits per NFPA 79 §13.2.

Motor leads (U/V/W) and AC mains.

Equipment grounding per NFPA 79.

Originally spec'd for VFD jam signal +24V; per as-built note, not currently installed (review).

Pushbutton common +24V supply protection (in EURO S4LH).

| Item / Function                            | Manufacturer       | Model / Part Number  |
|--------------------------------------------|--------------------|----------------------|
| Motor PE145TC-2-4                          | Various            | PE145TC-2-4          |
| Circuit Breaker NBE7                       | Various            | NBE7                 |
| 30 A Branch Fuse (TBD)                     | Various            | 30 A fuse            |
| DC Power Supply 1866-1441-ND               | Digi-Key vendor    | 1866-1441-ND         |
| VEVOR 3HP 120V VFD                         | VEVOR              | VEVOR 3HP 120V VFD   |
| ATO 3HP 120V VFD                           | ATO                | ATO 3HP 120V VFD     |
| 110V 20A 3HP VFD (generic)                 | Generic            | 110V 20A 3HP VFD     |
| WEG Single Phase Motor                     | WEG                | WEG Single Phase 3HP |
| Worldwide Electric NT3-18-213T             | Worldwide Electric | NT3-18-213T          |
| VEVOR High Torque Motor                    | VEVOR              | VEVOR 5HP 230V       |
| TECO L510 VFD                              | TECO               | L510 (2HP and 3HP)   |
| Marathon G0532 Gearmotor                   | Marathon           | G0532                |
| KB Electronics KBWC-15K                    | KB Electronics     | KBWC-15K             |
| Capacitive Touch Sensor TTP223B            | Various            | TTP223B              |
| Capacitive Touch Sensor CAP1293            | Microchip          | CAP1293-1-SN         |
| Capacitive Proximity Sensor 1864-1032-ND   | various            | 1864-1032-ND         |
| Magnetic Non-Contact Safety Sensor F3S-TGR | Omron              | F3S-TGR-NMPC-21-10   |
| RFID Safety Sensor RSS 36-I2-D             | Schmersal          | RSS 36-I2-D-R-ST5    |
| Safety Light Curtain F3SG-SR / PG          | Omron              | F3SG-SR / PG         |
| Ichor-provided Red/Green Lights            | Ichor (sponsor)    | Ichor-provided       |
| Ichor-provided E-stop                      | Ichor (sponsor)    | Ichor-provided       |

| From BOM File                | Reason Not Used                                                                          |
|------------------------------|------------------------------------------------------------------------------------------|
| BOM_02_20_26                 | Not used - GE 5KE182BC205B used instead for bench testing                                |
| BOM_1_29 / BOM_02_20_26      | Never purchased; Phoenix Contact TMC 8 3C 15A used instead                               |
| BOM_1_29                     | Never needed in final design                                                             |
| BOM_1_29                     | Superceded by Mean Well NDR-480-24                                                       |
| VFD_120V_3HP_Decision_Matrix | Rejected during selection (hobby-grade, limited documentation)                           |
| VFD_120V_3HP_Decision_Matrix | Too expensive (\$450-500) for project budget                                             |
| VFD_120V_3HP_Decision_Matrix | Considered but Huanyang HY02D211B-T selected                                             |
| Motor Spec                   | Considered but GE 5KE182BC205B used for bench                                            |
| Motor Spec                   | Considered but not purchased                                                             |
| Motor Spec                   | Considered but not purchased                                                             |
| Motor Spec                   | Considered but Huanyang selected                                                         |
| Motor Spec                   | Considered but not purchased                                                             |
| Motor Spec                   | Considered (triac speed control) but VFD approach used instead                           |
| BOM_1_30_beta (safety draft) | Does not meet industrial standards                                                       |
| BOM_1_30_beta (safety draft) | Chip only, needs interface board                                                         |
| BOM_1_30_beta (safety draft) | Too expensive (\$109.25)                                                                 |
| BOM_1_30_beta (safety draft) | Too expensive; out of stock at retailers                                                 |
| BOM_1_30_beta (safety draft) | Suspected non-retail / industrial-only                                                   |
| BOM_1_30_beta (safety draft) | Out of scope for capstone budget                                                         |
| BOM_02_20_26                 | Not applicable - Wamco 525 used as fault light; button LEDs provide ready/run indication |
| BOM_02_20_26                 | Replaced with Cutler-Hammer/Eaton E22B1 procured by team                                 |



| Option                                       | Price Score<br>(50%) | Current<br>Capability (20%) | Build Quality<br>(15%) |
|----------------------------------------------|----------------------|-----------------------------|------------------------|
| VEVOR 3HP 120V VFD                           | 9                    | 6                           | 5                      |
| 110V 20A 3HP VFD (generic)                   | 8                    | 7                           | 6                      |
| Huanyang 3HP VFD (HY02D211B-T) -<br>SELECTED | 7                    | 7                           | 6                      |
| ATO 3HP 120V VFD                             | 5                    | 9                           | 8                      |

| Documentation /<br>Support (10%) | Brand /<br>Reliability (5%) | Total Score |
|----------------------------------|-----------------------------|-------------|
| 4                                | 4                           | 7.05        |
| 5                                | 5                           | 7.05        |
| 5                                | 5                           | 6.55        |
| 7                                | 7                           | 6.55        |

## Notes

Lowest cost (\$70-95). Hobby-grade, limited docs.

Best balance (\$130-150). Good for 20A circuits.

Common CNC choice (\$120-200). Selected for project.

Higher quality but too expensive (\$450-500).